The First Model Organism in Genetics – Garden Peas (*Pisum sativum*) – Seven characters studied by Mendel.

Character	Gene	Dominant Trait	Recessive Trait
Plant height	Le	Tall	Dwarf
Pod shape	V	Inflated	Constricted
Seed shape	R	Round	Wrinkled
Cotyledon colour	I	Yellow	Green
Flower position	Fa	Axial	Terminal
Flower colour	A	Purple	White
Pod colour	GP	Green	Yellow

Figure 2.3: Seven characters of *Pisum sativum* studied by Mendel.

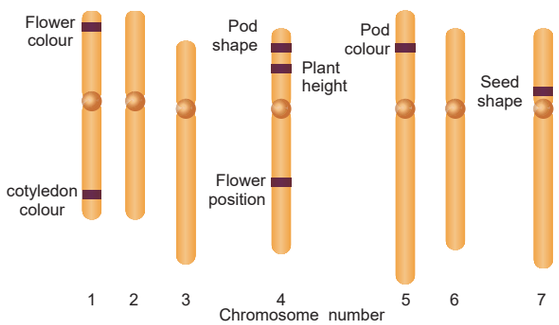


Figure 2.4: Mendel's seven characters in Garden Peas, shown on the plant's seven chromosomes

Mendel worked at the rules of inheritance and arrived at the correct mechanism before any knowledge of cellular mechanism, DNA, genes, chromosomes became available. Mendel insights and meticulous work into the mechanism of inheritance played an important role which led to the development of improved crop varieties and a revolution in crop hybridization.

Mendel died in 1884. In 1900 the work of Mendel's experiments were rediscovered by three biologists, **Hugo de Vries** of Holland, **Carl Correns** of Germany and **Erich von Tschermak** of Austria.

Can you identify Mendel's gene for **regulating white flower colour** in peas? Let us find the **molecular answer** to understand the gene function. Now the **genetic mystery of Mendel's white flowers is solved**.



Figure 2.5: Purple flower of Pea with Pea Gene A and White flower of Pea

It is quite fascinating to trace the Mendel's genes. In 2010, the gene responsible for regulating flower colour in peas were identified by an international team of researchers. It was called **Pea Gene A** which encodes a **protein** that functions as a transcription factor which is responsible for the production of **anthocyanin pigment**. So the flowers are purple. Pea plants with white flowers do not have anthocyanin, even though they have the gene that encodes the enzyme involved in anthocyanin synthesis.

Researchers delivered normal copies of gene A into the cells of the petals of white flowers by the gene gun method. When Gene A entered in a small percentage of cells of white flowers it is expressed in those particular cells, accumulated anthocyanin pigments and became purple.

In white flowers the gene A sequence showed a single-nucleotide change that makes the transcription factor inactive. So the mutant form of gene A do not accumulate anthocyanin and hence they are white.

2.2.3 Terminology related to Mendelism

Mendel noticed two different expressions of a trait – Example: Tall and dwarf. Traits are expressed in different ways due to the fact that a gene can exist in alternate forms (versions) for the same trait is called **alleles.**

If an individual has two identical alleles of

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